Hospital-Demo compared with the product requirements document

Reviewed 21 October 2026. Code: the Hospital-Demo repository, latest version on the main branch. Tested with no AI key, so the drafts came from the built-in sample mode, not from the real AI model.

Result in one paragraph

The code is not an exact match for the document. All 16 modules exist as a screen with working routes, but only 4 of them do everything the document says for them: the safety checks, the claim pack, the hospital dashboard and the patient timeline. The biggest cause is that, without an AI key, the drafts are pre-written text and do not come from what the doctor actually said.

Scorecard

What I checked	Count
Lines of the document that I could test (38 in total)	
Works as the document says	13
Works, but only on stored demo data	1
Partly works	15
Not built	6
Could not test (needs a working AI key)	3

What is wrong and how to fix it

Problem	What I saw	How to fix it
Drafts are pre-written without an AI key	A conversation about asthma still produced a leg-infection note. On an empty record the admission note, progress note and referral still invented a patient history.	Build drafts only from the words in the transcript and the record; leave a section empty when nothing supports it.
Investigation request form crashes	Returns a server error for the demo patient, so step 10 of the demo path cannot run.	Read the allergy details correctly (one small change) and return a clear error instead of crashing.
Uploaded lab report is ignored	A report with creatinine 1.0 came back as the stored value 3.1.	Read the text inside the uploaded PDF; if there is no text, ask the user to type the values in.
A discharged record is not locked	After discharge, a new note and a new task were still accepted. An empty discharge summary was accepted on an empty record.	Refuse new clinical entries after discharge. Check required documents before discharge and ask for a written reason to override open safety alerts.
Overdue tasks do not escalate by themselves	A task 6 hours late stayed unescalated until someone pressed the manual button. Tasks could jump from created straight to completed.	Run the escalation check automatically every minute. Enforce the order: assigned, acknowledged, in progress, completed.
Safety alerts pile up and never close	A new kidney result created a second alert for the same problem. Stopping the drug left the alert open. The allergy-contradiction check only fires on stored demo data.	One alert per problem, updated when new results arrive; close it automatically when the problem is gone. Record why each alert was dismissed and show how often alerts were useful.
No way to order, stop or give a medicine	The system cannot record that a drug was stopped or given, so 'a prescription is not an administration' cannot be shown.	Add records for starting, stopping and giving a medicine, each by the right person.

Problem	What I saw	How to fix it
Medicine comparison cannot be resolved item by item	Only the whole list can be approved.	Let the pharmacist decide on each difference (continue, hold, stop, change).
Documents and revenue checks are narrow	Only lab reports are read. Billing is compared only against procedures, using a text match.	Read prescriptions, radiology and discharge reports. Compare medicines, tests, consults and bed-days against the bill by code.
No separate views for nurse, pharmacist, billing, management	One fixed user does every action. Only the demo patient has records.	Add sign-in roles with permissions and a landing page for each. Give the other four patients real records.
No connection to hospital systems	No way to receive or send data. Everything is stored demo data.	Add receiving and sending in the standard hospital data format, with retries and a visible 'last synced' notice.
No AI credit counting	Usage is logged but not counted against a pack.	Count credits per operation using the document's schedule (report reading 1, voice note 2, medicine comparison 2, report analysis 4, discharge summary 5).

What already works

Nothing is saved to the patient record until a person approves it. Patient consent is required before a consultation note is signed. An allergy must be confirmed by a person before an admission note is approved. The progress note will not be signed until a clinician writes the examination and plan. The safety rules are fixed rules with evidence attached; the AI only words the message. Medicine comparison checks dose, route, frequency and reason. A new lab result flowed into the safety checks, medicine comparison, discharge summary, referral and timeline without extra work.

What I could not test

The real AI model and speech-to-text, because the only key available was rejected. That affects reading audio, drafting a note from a conversation with the real model, and reading real scans. The real model may do better than the sample mode I tested. I did not open the screens in a browser; I tested the server and the code, and the front end builds without errors. No hospital system connections exist yet, so there was nothing to test there.

Where the evidence is

Full report with file and line references: docs/PRD-gap-report.md. Twelve scripts that repeat every test: docs/prd-probes/. Both are in the Hospital-Demo-ARD folder. Next step: fix these on a separate branch, with automatic tests for each fix.